

09_TESTING.md

TASK 09

TESTING & QUALITY ASSURANCE

Status

Mandatory

Owner

QA & Security Lead

Supporting Guides

Project Director

Security Architect

Backend Lead

Frontend Lead

Principal Engineer

Estimated Time

3 to 5 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

09_QA_SECURITY_LEAD.md

06_BACKEND.md

07_FRONTEND.md

08_SECURITY.md

Mission

Verify that every feature of the Secure Dance Academy Management System works correctly, securely and reliably.

Testing should provide measurable confidence that the application is ready for deployment.

Every discovered issue should be documented, fixed, retested and verified.

Objectives

Validate every functional requirement.

Verify every security control.

Confirm system stability.

Measure performance.

Validate accessibility.

Produce complete testing evidence.

Testing Philosophy

Testing proves quality.

Security testing proves resilience.

Every important workflow should be verified.

Never assume software works because it compiles.

Evidence is required for every claim.

Testing Strategy

Apply the Testing Pyramid.

Static Analysis

↓

Unit Testing

↓

Integration Testing

↓

API Testing

↓

End-to-End Testing

↓

Security Testing

↓

Performance Testing

↓

Accessibility Testing

↓

Regression Testing

Every layer increases confidence.

Unit Testing

Use

Jest

Test

Utilities

Validation

Business Logic

Permission Checks

Service Layer

Repository Layer

Edge Cases

Error Handling

Target Coverage

Minimum 80%

Critical Modules

Minimum 90%

Integration Testing

Verify

Authentication

Authorization

Database Operations

Prisma

Repositories

Transactions

Audit Logging

Notifications

Module Communication

Test realistic workflows.

API Testing

Use

Postman

Test every endpoint.

Verify

Status Codes

Authentication

Authorization

Validation

Request Payloads

Responses

Pagination

Filtering

Sorting

Error Messages

Headers

Rate Limiting

Create reusable Postman collections.

End-to-End Testing

Use

Playwright

Test complete user journeys.

Administrator

Login

User Management

Reports

Audit Logs

Coach

Attendance

Performance

Activities

Parent

Children

Attendance

Notifications

Artist

Dashboard

Performance

Profile

Medical Records

Test using real browser interactions.

Security Testing

Use

OWASP ZAP

Perform

Spider Scan

Passive Scan

Active Scan

Header Analysis

Cookie Analysis

Authentication Testing

Authorization Testing

Session Testing

Input Validation Testing

Document every finding.

Static Code Analysis

Use

SonarQube

Review

Maintainability

Reliability

Security Hotspots

Duplicated Code

Complexity

Technical Debt

Resolve all Critical and High issues.

Performance Testing

Measure

Application Startup

Page Load Time

Dashboard Load Time

API Response Time

Database Queries

Search Performance

Pagination

Filtering

Memory Usage

Record benchmark results.

Accessibility Testing

Verify

WCAG 2.2 AA

Keyboard Navigation

Screen Reader Support

Colour Contrast

ARIA Labels

Semantic HTML

Responsive Layout

Focus Order

Accessibility failures must be corrected.

Browser Testing

Verify

Chrome

Edge

Firefox

Safari

Responsive behaviour should remain consistent.

Mobile Testing

Test

Responsive Layout

Touch Navigation

Forms

Tables

Authentication

Performance

Notifications

Every feature should remain usable.

Regression Testing

Repeat tests after every major change.

Authentication

Attendance

Medical Records

Reports

Dashboards

Performance

Settings

Regression protects existing functionality.

Bug Management

Every defect must include

Bug ID

Title

Description

Severity

Priority

Environment

Steps to Reproduce

Expected Behaviour

Actual Behaviour

Evidence

Suggested Fix

Verification Result

Severity Classification

Critical

Application unusable.

High

Major functionality or security issue.

Medium

Partial functionality affected.

Low

Minor usability or cosmetic issue.

Classify consistently.

Test Evidence

Collect

Screenshots

Console Logs

Test Reports

Coverage Reports

Performance Results

OWASP Reports

SonarQube Reports

Bug Reports

Evidence strengthens academic assessment.

Test Reports

Produce

Unit Test Report

Integration Test Report

API Test Report

Playwright Report

Performance Report

Accessibility Report

OWASP ZAP Report

SonarQube Report

Bug Summary

Regression Report

Quality Dashboard

Risk Review

Review

Known Risks

Resolved Risks

Residual Risks

Future Risks

Update mitigation status.

Quality Gates

Release only if

No Critical Defects

No High Security Risks

Tests Pass

Coverage Target Met

Performance Acceptable

Accessibility Passes

Documentation Updated

Architecture Unchanged

Deliverables

Master Test Plan

Test Cases

Test Scripts

Jest Suite

Postman Collection

Playwright Suite

OWASP ZAP Report

SonarQube Report

Bug Register

Regression Report

Performance Report

Accessibility Report

Coverage Report

Final QA Report

Acceptance Criteria

Every functional requirement passes.

Authentication passes.

Authorization passes.

API tests pass.

Security tests pass.

Performance targets are achieved.

Accessibility standards are met.

No unresolved Critical defects remain.

Update Project Memory

Mark

Testing

Completed

Record

Coverage Percentage

Defects Found

Defects Fixed

Remaining Risks

Performance Results

Accessibility Status

Next Task

Formal Methods Verification

Testing Review Checklist

Verify

Unit tests pass.

Integration tests pass.

API tests pass.

End-to-end tests pass.

Security scans are complete.

Static analysis passes.

Performance is acceptable.

Accessibility passes.

Regression passes.

Documentation is complete.

Definition of Done

The application has been comprehensively tested.

Every major workflow has been verified.

Security testing demonstrates strong protection.

Performance meets project goals.

Accessibility complies with WCAG 2.2 AA.

Testing evidence is complete and suitable for inclusion in the coursework report.

The application is approved for formal verification and final documentation.